Calculus II (MATH 162)

Final Exam (Fall,2025)

Department : Id number : Name :

풀이 과정을 자세히 기술해야 합니다.

1. (10pts) (a) Evaluate $\int_C (y+z)\,dx + (x+z)\,dy + (x+y)\,dz$, where C consists of the line segments from $(0,0,0)$ to $(1,0,1)$ and from $(1,0,1)$ to $(0,1,2)$.

(b) Evaluate $\int_C (x^3 \sin x + y^3)\,dx + (e^y \cos y - x^3)\,dy$, where C is the circle $x^2 + y^2 = 4$.

2. (10pts) (a) Evaluate $\iint_S x\,dS$, where S is the surface with vector equation $\mathbf{r}(u,v) = \langle u\cos v,\ u\sin v,\ v\rangle,\ 0 \le u \le 1,\ 0 \le v \le \pi/2$.

(b) Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where

$$\mathbf{F}(x,y,z) = (y^2+2x)\mathbf{i} + (x^3-2yz)\mathbf{j} + (z^2-y)\mathbf{k}$$

and S is the surface of the solid bounded by cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = y+1$.

Department : Id number : Name :

3. (10pts) Evaluate $\iint_R e^{xy}\,dA$, where R is the region enclosed by the curves $y=\sqrt{x}$ and $y=2\sqrt{x}$ and the hyperbolas $y=\frac{1}{x}$ and $y=\frac{4}{x}$.

4. (10pts) Find the work done by the force field

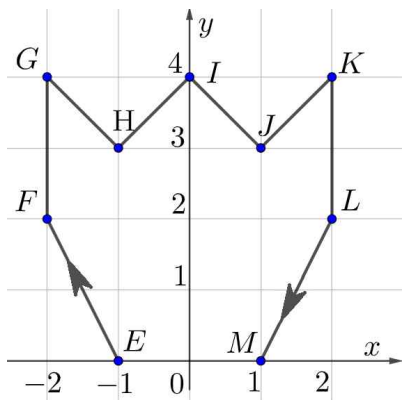
$$\mathbf{F}(x,y,z)=\frac{y}{1+x^2y^2}\,\mathbf{i}+\left(\frac{x}{1+x^2y^2}+z\sec^2\left(\frac{yz}{3}\right)\right)\mathbf{j}+y\sec^2\left(\frac{yz}{3}\right)\mathbf{k}$$

along the curve C given by

$$\mathbf{r}(t)=\left(\frac{t}{3}\cos\frac{\pi t}{3}\right)\mathbf{i}+\left(\frac{t}{3}\sin\frac{\pi t}{6}\right)\mathbf{j}+\pi\cos(\pi t)\mathbf{k},\ 0\le t\le 3.$$

Department : Id number : Name :

5. (10pts) Evaluate $\int_C \frac{y\,dx + x\left(x^2+y^2-1\right)dy}{x^2+y^2}$ if C is the curve consisting of the line segments $E \to F \to \ldots \to M$ in the figure.



6. (10pts) Use a parametric equation to find the surface area of S, where S is the surface given in spherical coordinates by $\rho = 1 - \cos\phi$. (cardioid revolution)

Department : Id number : Name :

7. (10pts) Let

$\mathbf{F}=\left(\ln(1+y^2)-x\,\mathrm{sech}(2z)\right)\mathbf{i}+\left(y-2\tan^{-1}(x)\right)\mathbf{j}+\tan^{-1}(\tanh z)\mathbf{k}$

Let S be the surface of the solid bounded by the half-cylinder $x^2+z^2=1\ (z\geq 0)$ and half-cylinder $y^2+z^2=1\ (z\geq 0)$ and the plane $z=0$. Find the flux of $\mathbf{F}$ across S.

8. (10pts) Let

$\mathbf{F}(x,y,z)=\left(ze^x\sin y+yz\right)\mathbf{i}+\left(ze^x\cos y+xy^2\right)\mathbf{j}+\left(e^x\sin y+xy\right)\mathbf{k}$

and let C be the curve of intersection of the paraboloid $z=x^2+y^2$ and the parabolic cylinder $z=1-x^2$, oriented counterclockwise as view from above.

(1) Find the parametric equation of C.

(2) Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$.

Department : Id number : Name :

9. (10pts) Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where

$$\mathbf{F}(x,y,z) = x^2 \sin z\,\mathbf{i} + y^2\mathbf{j} + (xy + x^2\cos z)\mathbf{k}$$

and S is the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = -2x + 3$, oriented downward.

10. (10pts) (a) 크레머 공식(Cramer's rule)을 이용하여 다음 연립방정식의 해에서 y의 값을 구하시오.

$$\begin{cases} x - y + w = 2 \\ 3x - z + w = 1 \\ 3y - 5z - 7w = 5 \\ -4x + y + z - w = 2 \end{cases}$$

(b) $A = \begin{pmatrix} a & b & c \\ \alpha & \beta & \gamma \\ x & y & z \end{pmatrix}$, $B = \begin{pmatrix} 2\alpha & 2\beta & 2\gamma \\ -x+5a & -y+5b & -z+5c \\ a-3\alpha & b-3\beta & c-3\gamma \end{pmatrix}$ 이고 $\det((2A)^T) = 24$ 일 때, 행렬식의 성질을 이용하여 $\det(-B^T A^{-1} B)$를 구하시오.

(c) $A = \begin{pmatrix} 2 & 0 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 2 & 1 & -6 \\ 0 & -1 & 2 \\ 5 & 3 & 7 \end{pmatrix}$ 이고 $D = AB - C^T$일 때, D행렬의 3행 3열의 요소(원소)를 구하시오.